Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
2 P-only DeepONet10	**10k**	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	**long**	2.20e-03	0.0192	0.0875	0.165	0.12	0.0203	6.87e-03	0.28	0.242	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.78e-03	0.0124	0.107	0.166	0.0455	0.027	3.34e-03	0.103	0.0928	0	Complete
4 T/P DeepONet10	**10k**	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	**long**	0.46	0.432	9.77	0.918	0.202	4.85	0.174	0.81	31	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.043	0.0234	0.653	0.243	0.139	0.883	0.0315	0.353	0.361	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.699	0.483	13.2	2.15	0.142	8.55	0.319	28.5	541	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.266	0.0204	3.64	0.0449	0.0177	5.71	0.203	1.04	1.61	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.